This page is the record for studying linear algebra, Hoffman & Kunze.

⌈Assume $T: V \to V$, V is finite dimensional vector space.

⌈Lemma. Let $\alpha \in V$ be an eigenvector, $T\alpha = c\alpha$.
For any $f \in F[x]$, $f(T)\alpha = f(c)\alpha$.

⌈Thm. Let $c_1, \cdots, c_k \in F$ be distinct eigenvalues. Then,

$$\dim \sum_{i=1}^{k} E_{c_i} = \sum_{i=1}^{k} \dim E_{c_i}$$

Moreover, if β_i $(1 \le i \le k)$ is basis for E_{c_i}, then $\bigcup_{i=1}^{k} \beta_i$ is basis for $\sum_{i=1}^{k} E_{c_i}$

Proof. Since $W_1 + W_2 \le V$ if $W_1, W_2 \le V$ and $\langle \beta_1 \rangle + \langle \beta_2 \rangle = \langle \beta_1 \cup \beta_2 \rangle$

$$\sum_{i=1}^{k} E_{c_i} = \sum_{i=1}^{k} \mathrm{span}(\beta_i) = \mathrm{span}\left(\bigcup_{i=1}^{k} \beta_i \right)$$

Meanwhile, for each $1 \le i \le k$, let $B_i \in E_{c_i}$. Suppose that

$$B_1 + B_2 + \cdots + B_k = 0.$$

For any polynomial $f \in F[x]$,

$$0 = f(T)(0) = f(T)(B_1 + \cdots + B_k) = f(T)(B_1) + \cdots + f(T)(B_k) = f(c_1)B_1 + \cdots + f(c_k)B_k$$

Using Lagrange's interpolation, choose f_i $(1 \le i \le k)$ s.t.

$$f_i(c_j) = \delta_{ij}.$$

Now, $0 = \sum_{i=1}^{k} f_j(c_i) B_i = B_j$, for each $1 \le j \le k$. In summary,

$$B_1 + \cdots + B_k = 0 \quad (B_i \in E_{c_i}) \Rightarrow B_i = 0 \quad (1 \le i \le k)$$

Now, the set $\bigcup_{i=1}^{k} \beta_i$ is linearly independent, because

if $\sum_{i=1}^{k} \sum_{j=1}^{r_i} a_{i,j} \cdot B_{i,j} = 0$ $(B_{i,j} \in \beta_i)$, then $\sum_{j=1}^{r_i} a_{i,j} B_{i,j} = 0$ $(1 \le i \le k)$,

thus all $a_{i,j} = 0$. Now proved.

⌈Thm. Let $T: V \to V$ be a linear operator on a finite-dimensional space V.
Let $c_1, \cdots, c_k$ be the distinct eigenvalues of T. TFAE:

a). T is diagonalizable.

b). $ch(T) = \prod_{i=1}^{k} (x - c_i)^{d_i}$ where $d_i = \dim E_{c_i}$.

c). $\dim V = \sum_{i=1}^{k} \dim E_{c_i}$

「Minimal Polynomial.

「Thm. The charaterstic and minimal polynomials for T have the same roots.

Proof. Let p be the minimal polynomial for T.

Suppose that $c \in F$ is root of p. That is, $p(c)=0$. Then,

$$p = (x-c)\cdot q \quad (\deg q < \deg p)$$

for some $q \in F[x]$. Since p is the minimal, $q(T) \neq 0$. Take $\beta \in V$ s.t. $q(T)(\beta) \neq 0$.
Then, $0 = p(T)(\beta) = (T-cI)\cdot q(T)(\beta)$. Thus $T \circ q(T)(\beta) = c\beta$, c is eigenvalue.
Conversely, let $c \in F$ be an eigenvalue of T. That is, $T\beta = c\beta$ for some nonzero $\beta \in V$.
Now, $0 = p(T)(\beta) = p(c)\beta$ and since β is nonzero, $p(c)=0$. ▨

Note; the Minimal Polynomial is Not necessarily irreducible; actually,

$$e_T : F[x] \to \mathcal{L}(V,V) ; f(x) \mapsto f(T)$$

the minimal polymical is the generator of the ideal $\ker e_T$,
where e_T is the ring-Homomorphism. Since $\mathcal{L}(V,V)$ is a ring under $(+,\circ)$
and it is contains a zero divisor; for instance, $\dim V = 2$,

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 0$$

Thus, even if $p \in F[x]$ is the minimal polynomial,

$$0 = p(T) = f(T)\cdot g(T) \quad \text{for some } \deg f, \deg g < \deg p$$

can be happen.

「Invariant.

(199.P)

Let $T, U : V \to V$ be linear operators, and commute.

Both $\operatorname{Im} U$ and $\operatorname{Ker} U$ are invariant under T, because;

$y \in \operatorname{Im} U \Rightarrow y = U(x)$ for some $x \in V \Rightarrow T(y) = T(U(x)) = U(T(x)) \in \operatorname{Im} U$

$y \in \operatorname{Ker} U \Rightarrow U(y) = 0 \Rightarrow 0 = T(U(y)) = U(T(y)) \Rightarrow T(y) \in \operatorname{Ker} U$.

Since T commute with $g(T)$, where $g \in F[x]$,

$\operatorname{Ker} g(T)$ and $\operatorname{Im} g(T)$ are invariant under T.

「Lemma. Let $W \leq V$ be an invariant subspace for $T : V \to V$. Then,

$ch(T_W)$ divides $ch(T)$. Moreover, the minimal polynomial of T_W divides minimal polynomial of T

Proof. By the calculation,

$$[T]_\beta = \begin{bmatrix} [T_W]_{\beta'} & B \\ O & C \end{bmatrix}$$ where β and β' are basis for V and W, respectively, and $\beta' \subset \beta$

Now, $$ch(T) = \det(xI - [T]_\beta) = \det(xI - [T_W]_{\beta'}) \cdot \det(xI - C)$$
$$= ch(T_W) \cdot \det(xI - C),$$

thus the first assertion is proved.

「Conductor.

「Def. Let W be an invariant subspace for $T: V \to V$, and $\alpha \in V$ be a vector.
Define T-Conductor of α into W as;

$$S_T(\alpha; W) \overset{def}{=} \{g \in F[x] \mid g(T)(\alpha) \in W\}.$$

In particular, if $W = \{0\}$ the Conductor is called the T-annihilator of α.

「Lemma. The Conductor $S(\alpha; W)$ is an Ideal in $F[x]$.

Proof. First, if $T[W] \subseteq W$, then $g(T)[W] \subseteq W$, because for any $n \in \mathbb{N}$,

$$\beta \in W \Rightarrow T^n(\beta) = T^{n-1}(\underbrace{T(\beta)}_{\in W}) = \cdots = T(T^{n-1}(\beta)) \in W.$$

And since $W \leq V$, it is closed under the addition and Scalar multiplication, $g(T)[W] \subseteq W$. Now, to show $S(\alpha; W)$ is an Ideal, first show that it is space.
For any $f, g \in S(\alpha; W)$ and $c \in F$, $(cf+g)(T) = cf(T) + g(T)$, therefore

$$(cf+g)(T)(\alpha) = cf(T)(\alpha) + g(T)(\alpha) \in W$$

Thus it is subspace. Moreover, for any $f \in F[x]$ and any $g \in S(\alpha; T)$,

$$(fg)(T)(\alpha) = \underbrace{f(T)\underbrace{g(T)(\alpha)}_{\in W}}_{\in W} \in W.$$

Thus $fg \in S(\alpha; W)$. ∎

Note: Given the linear operator T and T-invariant subspace W, and any vector $\alpha \in V$,

$$\varphi: F[x] \to V/W : f(x) \mapsto f(T)(\alpha) + W$$

is $F[x]$-module Homomorphism, so $S_T(\alpha; W) = \ker \varphi$.

Lemma. If the minimal polynomial $P(x) \in F[x]$ for T factors as:

$$p = (x - c_1)^{r_1} \cdots (x - c_k)^{r_k}, \quad c_i \in F.$$

Condition

and let W be a proper subspace of V which is invariant under T,

then there is a vector $\alpha \in V$ s.t

results

1) $\alpha \notin W$

2) For some eigenvalue $c \in F$ of T, $(T - cI)(\alpha) \in W$.

Proof. Let $\beta \in V \setminus W$ be given. Let $g \in F[x]$ be the T-conductor of β into W. That is, $g(T)(\beta) \in W$. Since $(g(x)) = S_T(\alpha; W)$ and $P(T)(\alpha) = 0 \in W$, $P \in S_T(\alpha; W)$ therefore g divides p. And, since $\beta \notin W$, g must be Non Constant. Now,

$$g = (x - c_1)^{e_1} \cdots (x - c_k)^{e_k}, \quad 0 \le e_i \le r_i, \text{ at least one } e_i \text{ is positive.}$$

Let j be a integer s.t e_j is positive. Then,

$$g = (x - c_j) \cdot h \quad \text{for some } h \in F[x].$$

Since g is the generator of $S_T(\alpha; W)$, $h(T)\beta \notin W$. But,

$$(T - c_j I)(h(T)\beta) = g(T)(\beta) \in W.$$

Thm. Let V be a finite dimensional vector space over F, and $T: V \to V$ be a linear. T is Triangulable if and only if minimal polynomial of T splits over F.

Proof. If T can be represented by triangular, then clear, since: the determinant of triangle is product of diagonals. Conversely, suppose that

$$P(x) = (x - c_1)^{r_1} \cdots (x - c_k)^{r_k}$$

is the minimal polynomial of T. Since P shares the roots with $ch(T)$, c_i's are eigenvalues.

Choose $\alpha_1 \in V$ s.t $\alpha_1 \notin \{0\}$ and $(T - cI)(\alpha_1) \in \{0\}$, for some eigenvalue $c \in F$.

Choose $\alpha_2 \in V$ s.t $\alpha_2 \notin \langle \alpha_1 \rangle$ and $(T - cI)(\alpha_2) \in \langle \alpha_1 \rangle$, //

$$T(\alpha_2) = c_1\alpha_1 + c_2\alpha_2$$

Thm. Let V be a vector space over F.
For any non zero linear functional $f \in V^*$, $\ker f$ is hyperspace of V.
Conversely, every hyperspace in V is the kernel for some linear functional on V.
Proof. Let $f \in V^*$ be a non-zero linear functional. Since f is non zero,
choose $\alpha \in V \setminus \ker f$. (or $f(\alpha) \neq 0$). We shall show that: $V = N_f + \langle \alpha \rangle$.
Let $\beta \in V$ be given. Put $c = f(\beta)/f(\alpha)$. (It can be defined because $f(\alpha) \neq 0$).
Then, $f(\beta - c\alpha) = f(\beta) - cf(\alpha) = 0$, thus $\beta - c\alpha \in \ker f$, or $\beta \in \ker f + \langle \alpha \rangle$.

Conversely, let $N \leq V$ be the hyperspace. Let $\alpha \in V \setminus N$.
Since N is the hyperspace, $N + \langle \alpha \rangle = V$. Therefore, for each $\beta \in V$,

$$\beta = \gamma + c\alpha \quad \text{for some } \gamma \in N,\ c \in F.$$

This γ and c are uniquely determined, because $N \cap \langle \alpha \rangle = \{0\}$.
Define $g: V \to F$ as $g(\beta) = c$. Then, $\ker g = N$.

$$\begin{pmatrix} a_1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & a_3 \end{pmatrix}^2 = \begin{pmatrix} a_1^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & a_3^2 \end{pmatrix}$$

Lemma. Let V be a fin. dim v.s over F, and let $T: V \to V$ be a linear operator. Suppose that the minimal polynomial of T splits over F as:

$$p = (x-c_1)^{r_1} \cdot \cdots \cdot (x-c_k)^{r_k} \quad (c_i \in F)$$

Then, for any proper T-invariant subspace $W < V$, there exists a vector $\alpha \in V$ s.t

$\alpha \notin W$ and for some c_i, $(T - c_i I)(\alpha) \in W$.

Proof. Let $\beta \in V \setminus W$ be given, and let $g \in F[x]$ be a T-conductor of β into W. Then, g divides p. And, since $\beta \notin W$, g is not constant. Therefore,

$$g = (x-c_1)^{e_1} \cdots (x-c_k)^{e_k} \quad (0 \le e_i \le r_i)$$

and at least one of e_i is positive. Put j s.t $e_j > 0$. Then,

$$g = (x - c_j) \cdot h \qquad \text{where } h = g/(x-c_j) \in F[x].$$

Since g is T-conductor of β, $h(T)(\beta) \notin W$, but $(T - c_j I)(h(T)(\beta)) = g(T)(\beta) \in W$.

Thm. Let V be a finite dimensional vector space over F, and $T: V \to V$ be a linear. T is Triangulable if and only if minimal polynomial of T splits over F.

Proof. If T is triangulable with respects to a basis β, then the characteristic polynomial is

$$f(x) = \det([T]_\beta - xI) = \prod_{i=1}^{n} (A_{ii} - x) \quad \text{where } A_{ii} \text{ is } i\text{'th diagonal element of } [T]_\beta$$

thus, Proved. Conversely, choose $\alpha_1 \in V$ s.t $\alpha_1 \notin \{0\}$, $(T - c_1 I)(\alpha_1) \in \{0\}$ for some eigenvalue $c_1 \in F$ of T. Choose $\alpha_2 \in V$ s.t $\alpha_2 \notin \langle \alpha_1 \rangle$, $(T - c_2 I)(\alpha_2) \in \langle \alpha_1 \rangle$, it is possible because $\langle \alpha_1 \rangle$ is T-invariant. Again, choose $\alpha_3 \in V$ s.t $\alpha_3 \notin \langle \alpha_1, \alpha_2 \rangle$, $(T - c_3 I)(\alpha_3) \in \langle \alpha_1, \alpha_2 \rangle$. It is possible again, $T(\alpha_2) \in \langle \alpha_1, \alpha_2 \rangle$.
Now, we obtain $\{\alpha_1, \cdots, \alpha_n\}$ s.t

$$\begin{aligned}
T(\alpha_1) &= c_1 \alpha_1 \\
T(\alpha_2) &= a_{12} \alpha_1 + c_2 \alpha_2 \\
&\vdots \\
T(\alpha_n) &= a_{1n} \alpha_1 + a_{2n} \alpha_2 + \cdots + c_n \cdot \alpha_n .
\end{aligned}$$

⌈Thm. Let V be a fin. dim. v.s over F, and $T: V \to V$ be a linear operator.
T is diagonalizable if and only if minimal polynomial of T splits and separable.

Proof. Suppose that T is diagonalizable. Let $c_1, \cdots, c_k$ be distinct eigenvalues.
Then, for any eigenvector $\alpha \in V$, there is a $c_i \in F$ s.t. $(T - c_i I)(\alpha) = 0$.
Therefore, for $p = (x - c_1) \cdots (x - c_k)$,

$$p(T) = (T - c_1 I) \cdots (T - c_k I)(\alpha) = 0$$

because $\{T - c_1 I, \cdots, T - c_k I\}$ commute. Since T is diagonalizable,
for any vector of V can be represented by linear combination of eigenvectors,

$$p(T) = 0$$

Thus proved. Conversely, let W be a spanned subspace by all of eigenvectors of T.
Suppose that $W \neq V$. By the construction, W is T-invariant. Therefore, by the Lemma,
there is $\alpha \in V \setminus W$ and eigenvalue $c_j \in F$ s.t. $\beta = (T - c_j I)(\alpha) \in W$.
Since $\beta \in W$, $\beta = \beta_1 + \cdots + \beta_k$ where $T(\beta_i) = c_i \beta_i$, $1 \le i \le k$. Therefore, for any $h \in F[x]$

$$h(T)(\beta) = \sum_{i=1}^{k} h(c_i) \cdot \beta_i \in W.$$

Put $p = (x - c_j) g$. Also, $g - g(c_j) = (x - c_j) h$. Then,

$$g(T)(\alpha) - g(c_j)(\alpha) = h(T)(T - c_j I)(\alpha) = h(T)(\beta) \in W$$

And, $0 = p(T)(\alpha) = (T - c_j I) g(T)(\alpha) = g(T)(T - c_j I)(\alpha) \in W$.
Therefore $g(c_j)(\alpha) \in W$, and since $\alpha \notin W$, $g(c_j) = 0$. Contradicts to $p = (x - c_j)^2 \cdot g'$.

「Lemma. Let $f, g : V \to F$ be linear functionals on V.

$g = c \cdot f$ for some $c \in F$ if and only if $\ker f \leq \ker g$.

Proof. If $f = 0$, then trivial. Assume $f \neq 0$. Then, $\ker f$ is the hyperspace of V.
Let $\alpha \in V$ s.t $f(\alpha) \neq 0$. Put $c = g(\alpha)/f(\alpha)$. Then, $g - cf$ is a linear functional s.t
$\ker f \leq \ker(g - cf)$ and $\alpha \in \ker(g - cf)$. Thus $\ker(g - cf) = V$, or $g - cf = 0$.

「Thm. Let $g, f_1, \cdots, f_r$ be linear functionals on V. Then,

g is linear combination of $\{f_1, \cdots, f_r\}$ iff $\bigcap_{i=1}^{r} \ker f_i \leq \ker g$.

Proof. If $g = \sum_{i=1}^{r} c_i f_i$ for some $c_i \in F$, then for any $\alpha \in \bigcap_{i=1}^{r} \ker f_i$, $g(\alpha) = \sum_{i=1}^{r} c_i f_i(\alpha) = 0$.
Conversely, we shall use the Mathematical induction for $r \in \mathbb{N}$.
By the previous Lemma, if $r = 1$, it is proved. Suppose that it is held for $r = k-1$.
Let $f_1, \cdots, f_k$ be linear functionals on V s.t

$$\bigcap_{i=1}^{k} \ker f_i \leq \ker g.$$

Let $g, f_1, \cdots, f_{k-1}$ be restricted by $\ker f_k$. That is,

$$f_i|_{\ker f_k} : \ker f_k \to F ; \alpha \mapsto f_i(\alpha). \quad (i = 0, \cdots, k-1, \ f_0 = g)$$

And, let $\alpha \in \ker f_k$ s.t for all $1 \leq i \leq k-1$, $f_i(\alpha) = 0$.
That is, $\alpha \in \bigcap_{i=1}^{k} \ker f_i$ and therefore $g|_{\ker f}(\alpha) = 0$, by the hypothesis.

By the assumption of the induction,

$$g|_{\ker f_k} = \sum_{i=1}^{k-1} c_i \cdot f_i|_{\ker f_k} \quad \text{for some } c_i \in F$$

Put
$$h = g - \sum_{i=1}^{k-1} c_i \cdot f_i.$$
Now, for any $\alpha \in \ker f_k$, $h(\alpha) = 0$, or $\ker f_k \leq \ker h$.

The previous Lemma gives that $h = c_k \cdot f_k$ for some $c_k \in F$. Now, $g = \sum_{i=1}^{k} c_i \cdot f_i$.

「Dual.

$$\begin{tikzcd}
V \arrow[r, "\alpha_i \mapsto f_i"] \arrow[d, "T"'] & V^* \arrow[r] & V^{**} \arrow[d, "T^{tt}"] \\
W \arrow[r, "\beta_i \mapsto g_i"'] & W^* \arrow[r] \arrow[u, "T^t"'] & W^{**}
\end{tikzcd}$$

$\dim \ker f = \dim V - 1$ where $f \in V^*$.

$\ker f := \{\alpha \in V \mid f(\alpha) = 0\}$

$S^\circ := \{f \in V^* \mid f(\alpha) = 0 \ \ \forall \alpha \in S\}$.

$\bigcap_{i=1}^{r} \ker g_i \not\leq W$

$W = \ker f_{k+1} \cap \ker f_{k+2} \cap \cdots \cap \ker f_n.$

$= F^{n-k}$

$$\begin{matrix} A_{11}x_1 + \cdots + A_{1n}x_n = 0 \\ \vdots \\ A_{m1}x_1 + \cdots + A_{mn}x_n = 0 \end{matrix}$$

$W \leq \ker f \iff \bigcap \ker g_i \leq \ker f.$

For each $1 \leq i \leq m$, define

$$f_i : F^n \to F \ ; \ (x_i) \mapsto \sum_{j=1}^{n} A_{ij} x_j$$

$\ker f_1 \cap \ker f_2 \cap \cdots \cap \ker f_m$

「$W \leq V$, $\{g_1, \cdots, g_r\}$ is basis for W°.

$$W = \bigcap_{i=1}^{r} \ker g_i$$

Proof. $W \leq \bigcap \ker g_i$ is trivial, because $W \leq \ker g_i$ for all $g_i \in W^\circ$.

Conversely, let $\{f_1, \cdots, f_m\} \subseteq V^*$ s.t. $W = \ker f_1 \cap \cdots \cap \ker f_m$. Then, for each $1 \leq i \leq m$,

$W \leq \ker f_i \iff f_i \in W^\circ \iff f_i$ is linear combination of $\{g_1, \cdots, g_r\}$

$\iff \bigcap_{i=1}^{r} \ker g_i \leq \ker f_i$

Thus proved.

Consider a system of linear equation;

$$\begin{matrix} A_{11}x_1 & + \cdots & + A_{1n}x_n & = 0 \\ \vdots & \ddots & \vdots & \vdots \\ A_{m1}x_1 & + \cdots & + A_{mn}x_n & = 0 \end{matrix}$$

Define $f_i : F^n \to F : (x_j) \mapsto (A_{ij} \cdot x_j)$ for each $1 \leq i \leq m$.

We want to find a subspace; $\ker f_1 \cap \cdots \cap \ker f_m$. (has dimension $\leq n-m$)

1). If $m < n$, it has a non-trivial solution.

2). If $m = n$ and

Def. Let $W_i \le V$.
If $\alpha_1 + \cdots + \alpha_k = 0$ $(\alpha_i \in W_i) \Rightarrow \alpha_i = 0$,
Then W_i's are independent.

Lemma. Let $W = \sum W_i$. TFAE
1) W_i's are independent.
2) For each $2 \le j \le k$, $W_j \cap (W_1 + \cdots + W_{j-1}) = \{0\}$.
3) If β_i is basis for W_i, then $\beta_1 \cup \cdots \cup \beta_k$ is basis for W.

proof. 1) ⇒ 2)
Let $\alpha \in W_j \cap \sum_{i=1}^{j-1} W_i$. Then, $\alpha = \alpha_j = \alpha_1 + \cdots + \alpha_{j-1}$. Then

$$\alpha_1 + \cdots + \alpha_{j-1} - \alpha_j = 0 \Rightarrow \alpha_j = 0.$$

2) ⇒ 1). Suppose $\alpha_1 + \cdots + \alpha_k = 0$ $(\alpha_i \in W_i)$. If not all zero, let j be the largest integer st $\alpha_j \ne 0$. Then, $0 = \alpha_1 + \cdots + \alpha_j \Rightarrow \alpha_j = -\alpha_1 - \cdots - \alpha_{j-1} \in W_j \cap \{\sum_{i=1}^{j-1} W_i\}$.

Def. A linear operator $E: V \to V$ is called a projection if $E^2 = E$.

- $x \in \operatorname{Im} E \iff E(x) = x$

Proof. $x \in \operatorname{Im} E \Rightarrow x = E(\alpha)$ for some $\alpha \in V \Rightarrow E(x) = E^2(\alpha) = E(\alpha) = x$. Converse is trivial.

- $V = \operatorname{Im} E \oplus \operatorname{Ker} E$

Proof. Given $x \in V$, $x = E(x) + (x - E(x))$. Clearly $E(x) \in \operatorname{Im} E$, and

$$E(x - E(x)) = E(x) - E^2(x) = 0 \Rightarrow x - E(x) \in \operatorname{Ker} E$$

And, let $y \in \operatorname{Im} E \cap \operatorname{Ker} E$ be given. Then, $E(y) = y$ and $E(y) = 0$, thus $y = 0$.

Thm. Let $V = W_1 \oplus W_2$. Then, there is one and only one projection $E: V \to V$ s.t

$$\operatorname{Im} E = W_1 \quad \text{and} \quad \operatorname{Ker} E = W_2$$

Proof. Ask a toddler on the street.

Let E be a projection. For β_1, basis for $\operatorname{Im} E$, and β_2, basis for $\operatorname{Ker} E$,

$$[E]_{\beta_1 \cup \beta_2} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \quad \text{where } r = \operatorname{rank} E.$$

Thm. Let V be a fin. dim. V.S over F.

If $V = \bigoplus_{i=1}^{k} W_i$, then there exist k linear operators E_i on V s.t

$$E_i \circ E_j = \delta_{ij} \cdot E_j, \quad I = \sum_{i=1}^{k} E_i, \quad W_i = \operatorname{Im} E_i \quad (1 \le i \le k) \quad \cdots (*)$$

Conversely, if $E_1, \cdots, E_k$ are k linear operators on V which satisfy $(*)$, then

$$V = \bigoplus_{i=1}^{k} W_i$$

Proof. Suppose $V = \bigoplus_{i=1}^{k} W_i$. For each $1 \le j \le k$, define $E_j: V \to V; \sum_{i=1}^{k} \alpha_i \mapsto \alpha_j$. Proved.

Conversely, suppose that $E_1, \cdots, E_k$ are linear operators on V satisfying $(*)$.

Given $\alpha \in V$, $\alpha = E_1(\alpha) + \cdots + E_k(\alpha) \in W_1 + \cdots + W_k$. And, to show uniqueness such expression, let $\alpha = \alpha_1 + \cdots + \alpha_n$, where $\alpha_i \in W_i$. And, let $\alpha_i = E_i(\beta_i)$, for each $1 \le i \le k$.

Now, for each j, $1 \le j \le k$,

$$E_j(\alpha) = \sum_{i=1}^{k} E_j(\alpha_i) = \sum_{i=1}^{k} (E_j \circ E_i)(\beta_i) = E_j^2(\beta_j) = E_j(\beta_j) = \alpha_j$$

Thus Proved.

⌈Thm. Let V be a fin. dim. vs and $T: V \to V$ be a linear operator.
Let $E_1, \cdots, E_k$ be projection on V s.t $E_i \circ E_j = \delta_{ij} \cdot E_j$, $I = \sum_{i=1}^{k} E_i$.
Then, each $\operatorname{Im} E_i$ be invariant under T if and only if $T \circ E_i = E_i \circ T$, for all $1 \le i \le k$.
Proof. Suppose that T commutes with each E_i. Then, for any $\alpha \in \operatorname{Im} E_i$,

$$T(\alpha) = T(E_i(\alpha)) = E_i(T(\alpha)) \in \operatorname{Im} E_i.$$

Thus proved.

Conversely, suppose that each $\operatorname{Im} E_i$ is invariant under T. Given $\alpha \in V$, by the assumption,

$$\alpha = E_1(\alpha) + \cdots + E_k(\alpha) \Rightarrow T(\alpha) = T(E_1(\alpha)) + \cdots + T(E_k(\alpha))$$

And since each $\operatorname{Im} E_i$ is T-invariant, $T(E_i(\alpha)) = E_i(\beta_i)$ for some $\beta_i \in V$. Now,

$$(E_j \circ T \circ E_i)(\alpha) = (E_j \circ E_i)(\beta_i) = \delta_{ij} \cdot E_i(\beta_i)$$

Finally, $(E_j \circ T)(\alpha) = (E_j \circ T \circ E_1)(\alpha) + \cdots + (E_j \circ T \circ E_k)(\alpha) = E_j(\beta_j) = (T \circ E_j)(\alpha)$.

⌈Thm. Let T be a linear operator on fin. dim V.
If T is diagonalizable with distinct eigenvalues $c_1, \cdots, c_k$, then there are linear operators E_i,

$$T = c_1 E_1 + \cdots + c_k E_k,\quad I = E_1 + \cdots + E_k,\quad E_i \circ E_j = \delta_{ij} \cdot E_j,\quad \operatorname{Im} E_i = \ker(T - c_i I).$$

Conversely, if there are linear operators E_i and distinct scalars c_i satisfying the condition above, then T is diagonalizable and has c_i's as eigenvalue.
Proof. Suppose T is diagonalizable with distinct eigenvalues $c_1, \cdots, c_k \in F$. Then,

$$V = \ker(T - c_1 I) \oplus \cdots \oplus \ker(T - c_k I)$$

Define E_i be a projection onto $\ker(T - c_i I)$. Then, given $\alpha \in V$,

$$\alpha = E_1(\alpha) + \cdots + E_k(\alpha) \quad \text{and so}$$

$$T(\alpha) = T(E_1(\alpha)) + \cdots + T(E_k(\alpha)) = c_1 E_1(\alpha) + \cdots + c_k E_k(\alpha) = (c_1 E_1 + \cdots + c_k E_k)(\alpha).$$

Conversely,

⌈Thm. Primary decomposition Theorem.

Let T be a linear operator on fin. dim v.s V over a field F.
Let p be the minimal polynomial for T, with factorization

$$p = p_1^{r_1} \cdots p_k^{r_k} \quad (p_i \text{ are irreducibles, } r_i \in \mathbb{N}).$$

Then, for $W_i = \ker p_i(T)^{r_i}$ for each $(1 \le i \le k)$,

1). $V = W_1 \oplus \cdots \oplus W_k$

2). Each W_i is T-invariant

3). $T_i = T|_{W_i}$ has minimal polynomial as $p_i^{r_i}$.

Proof. For each $1 \le i \le k$, define $f_i = \prod_{j \ne i} p_j^{r_j}$. Since $p_1, \cdots, p_k$ are distinct prime elements, So $f_1, \cdots, f_k$ are relatively prime. That is,

$(f_1, \cdots, f_k) = (f_1) + \cdots + (f_k) = 1$. + More specifically, $(f_1, f_2, \cdots, f_m) = \left(\prod_{j>m}^{k} p_j^{r_j}\right)$

Now, there are polynomials $g_1, \cdots, g_k$ s.t

$$\sum_{i=1}^{k} f_i g_i = 1.$$

Moreover, $f_i \cdot f_j$ is divided by p if $i \ne j$. Let $E_i = (f_i \cdot g_i)(T)$, for each i. Then,

$$E_1 + \cdots + E_k = I \quad \text{and} \quad E_i \circ E_j = 0 \text{ if } i \ne j,$$

the second assertion is proved by

$$E_i \circ E_j = (f_i g_i)(T) \circ (f_j g_j)(T) = (g_i g_j)(T) \circ (f_i \cdot f_j)(T) = 0 \quad \text{if} \quad i \ne j.$$

Therefore, these $E_1, \cdots, E_k$ are projections. Moreover $\operatorname{Im} E_i = W_i = \ker p_i(T)^{r_i}$, because let $\alpha \in \operatorname{Im} E_i$. Then, $\alpha = E_i(\alpha)$, and so

$$p_i(T)^{r_i}(\alpha) = p_i(T)^{r_i} \circ E_i(\alpha) = p_i(T)^{r_i} \circ f_i(T) \circ g_i(T)(\alpha) = 0.$$

Conversely, let $\alpha \in \ker p_i(T)^{r_i}$. If $j \ne i$, then $f_j \cdot g_j$ is divided by $p_i^{r_i}$, and so

$$f_j(T) \circ g_j(T)(\alpha) = 0.$$

And thus, $E_i(\alpha) = \alpha$, $\alpha \in \operatorname{Im} E_i$.

「Inner Product. Let $F=\mathbb{R}$ or $F=\mathbb{C}$.

「Def. Inner product.

Let V be a vector space over F. A map

$$\langle \cdot,\cdot\rangle : V\times V \to F \,;\, (\alpha,\beta)\mapsto\langle\alpha,\beta\rangle$$

is called inner product if it satisfies the following: for any $\alpha,\beta,\gamma\in V$ and $c\in F$,

1) $\langle c\alpha+\beta,\gamma\rangle = c\langle\alpha,\gamma\rangle+\langle\beta,\gamma\rangle$

2). $\langle\alpha,\beta\rangle=\overline{\langle\beta,\alpha\rangle}$

3) If $\alpha\neq 0$, then $\langle\alpha,\alpha\rangle>0$

「Lemma. An orthogonal set of non-zero vectors is linearly independent.

Proof. Let $S \subseteq V \setminus \{0\}$ be an orthogonal set.
Suppose that $\alpha_1, \cdots, \alpha_n \in S$ are distinct vectors s.t

$$\beta = c_1\alpha_1 + \cdots + c_n\alpha_n \quad \text{for some } c_i \in F.$$

Then, for each $1 \le k \le n$,

$$\langle \beta, \alpha_k \rangle = \langle \sum c_i\alpha_i, \alpha_k \rangle = \sum c_i \langle \alpha_i, \alpha_k \rangle = c_k \cdot \langle \alpha_k, \alpha_k \rangle \Rightarrow c_k = \frac{\langle \beta, \alpha_k \rangle}{\langle \alpha_k, \alpha_k \rangle}$$

Since each $\alpha_k \neq 0$, if $\beta = 0$, then $c_1 = \cdots = c_n = 0$.
Furthermore, if $\beta \in V$ is linear combination of $\alpha_1, \cdots, \alpha_n \in S$,

$$\beta = \sum_{i=1}^{n} \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i .$$

「Gram-Schmidt Process.
Let V be a fin. dim. inn. prod. space, let $\{\beta_1, \cdots, \beta_n\} \subseteq V$ be a linearly independent subset.
Then, there exists an orthogonal subset $\{\alpha_1, \cdots, \alpha_n\} \subseteq V$ s.t

$$\mathrm{Span}\{\alpha_1, \cdots, \alpha_n\} = \mathrm{Span}\{\beta_1, \cdots, \beta_n\}.$$

Proof. Using mathematical induction for n. For $n=1$, take $\alpha_1 = \beta_1$, then clear.
Now, suppose that for $m \in \mathbb{N}$, the statement holds. Let $\{\beta_1, \cdots, \beta_{m+1}\}$ be lin. independent, and

$$\alpha_{m+1} = \beta_{m+1} - \sum_{i=1}^{m} \frac{\langle \beta_{m+1}, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i \quad \text{where } \alpha_1, \cdots, \alpha_m \text{ are obtained by hypothesis.}$$

Then $\alpha_{m+1} \neq 0$, because β_{m+1} is independent with $\{\alpha_1, \cdots, \alpha_m\}$. Moreover, for $1 \le j \le m$,

$$\langle \alpha_{m+1}, \alpha_j \rangle = \langle \beta_{m+1}, \alpha_j \rangle - \sum_{i=1}^{m} \frac{\langle \beta_{m+1}, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \langle \alpha_i, \alpha_j \rangle = \langle \beta_{m+1}, \alpha_j \rangle - \langle \beta_{m+1}, \alpha_j \rangle = 0.$$

Now, $\{\alpha_1, \cdots, \alpha_{m+1}\}$ is orthogonal, non-zero set, thus linearly independent by the Lemma.
Moreover, it is subset of $\mathrm{Span}\{\beta_1, \cdots, \beta_{m+1}\}$, the replacement Theorem gives result.

「Orthogonal Projection.

「Thm. Let W be a subspace of an inner prod. space V, and let $\beta \in V$.

1). A vector $\alpha \in W$ satisfies $\forall \gamma \in W$, $\|\beta-\alpha\| \le \|\beta-\gamma\|$ if and only if $\beta - \alpha \in W^\perp$.

2). If such vector in 1) exists, then it it is unique.

3). If W is finite-dimensional with an orthonormal basis $\{\alpha_1, \cdots, \alpha_n\}$ then

$$\alpha = \sum_{i=1}^{n} \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i.$$

is the unique best approximation.

Proof. Suppose that for all $\gamma \in W$, $\|\beta - \alpha\| \le \|\beta - \gamma\|$. Then,

$$0 \le \|\beta-\gamma\|^2 - \|\beta-\alpha\|^2 = \|\alpha-\gamma\|^2 + 2 \cdot \mathrm{Re}\langle \beta-\alpha, \alpha-\gamma\rangle \text{ for all } \gamma \in W.$$

Take for any $r \in W$, $r = \alpha - (\alpha - r)$, we can write

$$\|r\|^2 + 2\,\mathrm{Re}\langle \beta-\alpha, r\rangle \ge 0 \quad \text{for all } r \in W$$

Now, for any $\mu \in W \setminus \{\alpha\}$, take $r = \dfrac{-\langle \beta-\alpha, \alpha-\mu\rangle}{\|\alpha-\mu\|^2} \cdot (\alpha - \mu)$. Then,

$$\frac{|\langle \beta-\alpha, \alpha-\mu\rangle|^2}{\|\alpha-\mu\|^2} - 2 \cdot \frac{|\langle \beta-\alpha, \alpha-\mu\rangle|^2}{\|\alpha-\mu\|^2} \ge 0$$

it holds iff $\langle \beta-\alpha, \alpha-\mu\rangle = 0$. Therefore $\langle \beta-\alpha, \gamma\rangle = 0$ for all $\gamma \in W$.

Conversely, if $\beta - \alpha \in W^\perp$, then for any $\gamma \in W$,

$$\|\beta-\gamma\|^2 = \|\beta-\alpha\|^2 + \|\alpha-\gamma\|^2 + 2\cdot \mathrm{Re}\langle \beta-\alpha, \alpha-\gamma\rangle = \|\beta-\alpha\|^2 + \|\alpha-\gamma\|^2 \ge \|\beta-\alpha\|^2.$$

And, the uniqueness is given by: if $\alpha_1, \alpha_2 \in W$ satisfies $\beta-\alpha_1, \beta-\alpha_2 \in W^\perp$, then

for any $\gamma \in W$, $\langle \beta-\alpha_1, \gamma\rangle = \langle \beta - \alpha_2, \gamma\rangle = 0$

$$\Rightarrow \langle \alpha_1 - \alpha_2, \gamma\rangle = 0$$

Since $\alpha_1 - \alpha_2 \in W$, $\langle \alpha_1-\alpha_2, \alpha_1-\alpha_2\rangle = 0$, thus $\alpha_1 = \alpha_2$.

And, the last assertion is proved by: for each $1 \le k \le n$,

$\langle \beta-\alpha, \alpha_k\rangle = \langle \beta, \alpha_k\rangle - \langle \alpha, \alpha_k\rangle = 0$, therefore $\beta - \alpha \in W^\perp$.

$\boxtimes$

「Def. If for all $\beta \in V$, there is such $\alpha \in W$, then

$$E : V \to V ; \beta \mapsto \alpha$$

is orthogonal projection.

「Orthogonal Projection - Revised.

Let V be an inn. prod space, (not necessary finite dimensional) and $W \leq V$ be a subspace. For any vector $\beta \in V$, if $\alpha \in W$ satisfies that;

$$\|\beta - \alpha\| \leq \|\beta - \gamma\| \quad \text{for all } \gamma \in W \quad \cdots (*)$$

Then, the α is called best approximation to β by W, simply b. app.

「Thm. $\alpha \in W$ is b. app to $\beta \in V$ by W iff for all $\mu \in W$, $\langle \beta - \alpha, \mu \rangle = 0$.

Proof. For fixed $\alpha \in W$ and $\beta \in V$, for any $\gamma \in W$,

$$\|\beta - \gamma\|^2 = \|\beta - \alpha + \alpha - \gamma\|^2 = \|\beta - \alpha\|^2 + \|\alpha - \gamma\|^2 + 2 \cdot \mathrm{Re}\langle \beta - \alpha, \alpha - \gamma \rangle$$

Therefore, the condition $(*)$ holds if and only if

$$0 \leq \|\alpha - \gamma\|^2 + 2 \cdot \mathrm{Re}\langle \beta - \alpha, \alpha - \gamma \rangle \quad \text{for all } \gamma \in W.$$

Now take $\gamma = \alpha + \dfrac{\langle \beta - \alpha, \alpha - \mu \rangle}{\|\alpha - \mu\|^2} \cdot (\alpha - \mu)$ for any $\mu \in W \setminus \{\alpha\}$.

Then, $$0 \leq \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} - 2 \cdot \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} = -\frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2}$$

And the inequality holds iff $\langle \beta - \alpha, \alpha - \mu \rangle = 0$, and all nonzero vectors in W can be represented by $\alpha - \mu$, thus proved.

Conversely, if $\beta - \alpha$ is orthogonal to every vector in W, then $\mathrm{Re}\langle \beta - \alpha, \alpha - \gamma \rangle = 0$, thus $\|\beta - \gamma\|^2 - \|\beta - \alpha\|^2 = \|\alpha - \gamma\|^2 \geq 0$. ☒

「Thm. If b. app to $\beta \in V$ by W exists, then it is unique.

Proof. Suppose that $\alpha_1, \alpha_2 \in W$ satisfies for all $\gamma \in W$,

$$\langle \beta - \alpha_1, \gamma \rangle = \langle \beta - \alpha_2, \gamma \rangle = 0.$$

Then, take $\gamma = \alpha_1 - \alpha_2$, so $\langle \alpha_1 - \alpha_2, \alpha_1 - \alpha_2 \rangle = 0$. ☒

「Def. For all $\beta \in V$, there is b. app $\alpha \in W$, then the map

$$P : V \to V : \beta \mapsto \alpha$$

is called an orthogonal projection on W.

Thm. Let W be a fin. dim subspace of inn. prod space V. Then, the orthogonal projection $P: V \to V$ exists. Furthermore, P is idempotent linear operator with $\operatorname{Im} P = W$, $\ker P = W^{\perp}$.

Proof. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for W. Then, a map

$$P: V \to V ; \beta \mapsto \sum_{i=1}^{n} \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i$$

1). It maps β to orthogonal projection by W.

For each $1 \le j \le n$,

$$\left\langle \beta - \sum_{i=1}^{n} \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i , \alpha_j \right\rangle = \langle \beta, \alpha_j \rangle - \langle \beta, \alpha_j \rangle = 0$$

So, $\beta - P(\beta) \in W^{\perp}$ for all $\beta \in V$. (And, such a vector is unique, thus well-defined)

2). It is linear.

Let $\beta, \gamma \in V$ and $c \in F$ be given. Then,

$$P(c\beta + \gamma) = \sum \frac{\langle c\beta + \gamma, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i = c \cdot \sum \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i + \sum \frac{\langle \gamma, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i = cP(\beta) + P(\gamma)$$

3). It is idempotent.

Given $\beta \in V$, $P(\beta) \in W$. So, $P^2(\beta) = P(\beta)$.

4). $\operatorname{Im} P = W$, $\ker P = W^{\perp}$.

$\operatorname{Im} P \subseteq W$ is clear, and for any $\gamma \in W$, $P(\gamma) = \gamma \in \operatorname{Im} P$. And,

$$\beta \in \ker P \iff P(\beta) = 0 \iff \langle \beta, \alpha_i \rangle = 0 \text{ for all } 1 \le i \le n \iff \beta \in W^{\perp}.$$

Note: let $E: V \to V$ be a projection. In general, E can not be determined by it's image.

For example, let $V = \mathbb{R}^3$, $W_1 = \operatorname{Span}\{(1,0,0)\}$, $W_2 = \operatorname{Span}\{(0,1,0), (0,0,1)\}$, $W_3 = \operatorname{Span}\{(0,1,0), (1,1,1)\}$

Then, $V = W_1 \oplus W_2 = W_1 \oplus W_3$, but $W_2 \neq W_3$. Moreover,

$E: V \to V : (x_1, x_2, x_3) \mapsto (x_1, 0, 0)$, $E': V \to V ; (x_1, x_2, x_3) \mapsto (x_1 - x_3, 0, 0)$

Both E and E' are projections with $\operatorname{Im} E = \operatorname{Im} E' = \operatorname{Span}\{(1,0,0)\}$ but $E = E'$, $\ker E = W_2$ and $\ker E' = W_3$.

Corollary. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthogonal set of non-zero vectors in inn. prod space V. Then, for all $\beta \in V$, $\displaystyle\sum_{i=1}^{n} \frac{|\langle \beta, \alpha_i \rangle|^2}{\|\alpha_i\|^2} \le \|\beta\|^2$; the equality holds iff $\displaystyle\beta = \sum \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i$.

Proof. Take $\gamma = \sum [\langle \beta, \alpha_i \rangle / \|\alpha_i\|] \cdot \alpha_i$. Then, $\langle \beta - \gamma, \gamma \rangle = 0$. So,

$$\|\beta\|^2 = \|\gamma\|^2 + \|\beta - \gamma\|^2 \ge \|\gamma\|^2.$$

「Riesz representation Theorem.

「Thm. Let V be a fin. dim. inn. prod. v.s.
For any linear functional $f: V \to F$, there is a unique vector $\beta \in V$ s.t

$$\text{for all } \alpha \in V, \quad f(\alpha) = \langle \alpha, \beta \rangle$$

In other words, $\varphi: V \to V^*: \beta \mapsto \langle \cdot, \beta \rangle$ is a Conjugate isomorphism.

Proof. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for V.
Given linear functional $f: V \to F$, put

$$\beta = \sum_{j=1}^{n} \overline{f(\alpha_j)} \cdot \alpha_j .$$

then, for each $1 \leq k \leq n$,

$$\langle \alpha_k, \beta \rangle = \left\langle \alpha_k, \sum_{j=1}^{n} \overline{f(\alpha_j)}\, \alpha_j \right\rangle = \sum_{j=1}^{n} f(\alpha_j) \cdot \langle \alpha_k, \alpha_j \rangle = f(\alpha_k)$$

Clearly, $\langle \cdot, \beta \rangle$ is linear, thus the existence is proved. And, the uniqueness is shown;

if $\langle \alpha, \beta \rangle = \langle \alpha, \gamma \rangle$ for all $\alpha \in V$, then for $\alpha = \beta - \gamma$, $\langle \beta - \gamma, \beta - \gamma \rangle = 0$.

Thus, $\beta = \gamma$. ☒

Observe that a inn. prod space V with orthonormal basis $\{\alpha_1, \cdots, \alpha_n\}$.
Let $\alpha = x_1\alpha_1 + \cdots + x_n\alpha_n$ and $\beta = y_1\alpha_1 + \cdots + y_n\alpha_n$. Then,

$$\langle \alpha, \beta \rangle = x_1\overline{y_1} + \cdots + x_n\overline{y_n}$$

Meanwhile, for any linear functional $f \in V^*$,

$$f(\alpha) = f(\alpha_1)x_1 + \cdots + f(\alpha_n)x_n$$

If we want to β satisfies $\langle \alpha, \beta \rangle = f(\alpha)$, then take $y_i = \overline{f(\alpha_i)}$. That is,

$$\beta = \overline{f(\alpha_1)}\,\alpha_1 + \cdots + \overline{f(\alpha_n)} \cdot \alpha_n$$

「Adjoint.

「Def. Let T be a linear operator on inn. Prod. space V.
If there is a linear operator T^* s.t

$$\langle T(\alpha), \beta\rangle = \langle \alpha, T^*(\beta)\rangle \quad \text{for all } \alpha, \beta \in V$$

Then such T^* is called adjoint of T.

「Thm. If T is a linear operator on fin.dim inn. Prod space V, then T has a unique adjoint map T^*.

Prof. By the Riesz representation Thm, given $\beta \in V$, there is a unique $\beta' \in V$ s.t

$$\text{for all } \alpha \in V, \ \langle T(\alpha), \beta\rangle = \langle \alpha, \beta'\rangle.$$

Define $T^* : V \to V ; \beta \mapsto \beta'$. To show the linearity, let $\beta, \gamma \in V$ and $c \in F$ be given.
Then, for any $\alpha \in V$,

$$\langle \alpha, T^*(c\beta+\gamma)\rangle = \langle T(\alpha), c\beta+\gamma\rangle = \bar{c}\langle T(\alpha),\beta\rangle + \langle T(\alpha), \gamma\rangle = \bar{c}\langle \alpha, T^*(\beta)\rangle + \langle \alpha, T^*(\gamma)\rangle$$

$$= \langle \alpha, cT^*(\beta)\rangle + \langle \alpha, T^*(\gamma)\rangle = \langle \alpha, cT^*(\beta) + T^*(\gamma)\rangle.$$

So $T^*(c\beta+\gamma) = cT^*(\beta) + T^*(\gamma)$.

「Thm. Let V be a fin. dim. inn. Prod. space and let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis.
For any linear operator on V,

$$[T]_{\mathcal{B}} = \big(\langle T(\alpha_j), \alpha_i\rangle\big)_{i,j}.$$

Prof. For each $1 \le j \le n$,

$$T(\alpha_j) = \sum_{i=1}^{n} \langle T(\alpha_j), \alpha_i\rangle \cdot \alpha_i$$

Thus proved.

⌈Adjoint vs Transpose.

$T : V \to V$.

Adjoint: $T^* : V \to V : \beta \mapsto$

Transpose: $T^t : V^* \to V^* : g \mapsto g \circ T$

If V is fin. dim, then $V^* = \{\langle \cdot, \beta \rangle \mid \beta \in V\}$, so

$$g \circ T = \langle \cdot, \beta \rangle \circ T = \langle T(\cdot), \beta \rangle$$

s.t. $\langle T(\alpha), \beta \rangle = \langle \alpha, T^*(\beta) \rangle$ for all $\alpha, \beta \in V$

「Preserves Inner Product.

「Def. Let V and W be Inn. prod spaces over F.
We say that a linear map $T: V \to W$ preserves inner product iff for all $\alpha, \beta \in V$,

$$\langle T(\alpha), T(\beta) \rangle = \langle \alpha, \beta \rangle.$$

「Lemma Let V and W be Inn. Prod spaces over F with $\dim V = \dim W < \infty$.
For linear map $T: V \to W$, TFAE:
1). T preserves inner product.
2). T is isomorphism preserving inner product.
3). T maps every orthonormal basis for V onto orthonormal basis for W.
4). T maps a orthonormal //

Proof. 1) ⇒ 2) $\|T(\alpha)\|^2 = \|\alpha\|^2$. so $\ker T = \{0\}$. 2) ⇒ 1) is trivial.
2) ⇒ 3) is also trivial. If $\{\alpha_1, \cdots, \alpha_n\}$ is orthonormal basis for V, then $\{T(\alpha_1), \cdots, T(\alpha_n)\}$ is Too, $\langle T(\alpha_i), T(\alpha_j) \rangle = \langle \alpha_i, \alpha_j \rangle = \delta_{ij}$.
3) ⇒ 4) ㄱㄱ, 4) ⇒ 1) Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis with $\{T(\alpha_1), \cdots, T(\alpha_n)\}$ also orthonormal basis for W. Then, for any $\alpha, \beta \in V$, $\alpha = \sum x_i \alpha_i$, $\beta = \sum y_i \alpha_i$ f.s $x_i, y_i \in F$. Now,

$$\langle T(\alpha), T(\beta) \rangle = \langle \sum x_i \cdot T(\alpha_i), \sum y_j \cdot T(\alpha_j) \rangle = \sum_i \sum_j x_i \bar{y}_j \delta_{ij} = \sum_i x_i \bar{y}_i = \langle \alpha, \beta \rangle.$$

「Thm Let V and W be inn. Prod space over F, and let $T: V \to W$ be a linear map. Then,

T preserves inner product if and only if T preserves Norm.

Proof. One direction is trivial. Suppose that T preserves Norm. By the polarization identity,

$$\langle T(\alpha), T(\beta) \rangle = \frac{1}{4} \cdot \sum_{n=1}^{4} i^n \cdot \|T(\alpha) + i^n \cdot T(\beta)\|^2 = \frac{1}{4} \cdot \sum_{n=1}^{4} i^n \cdot \|T(\alpha + i^n \beta)\|^2 = \frac{1}{4} \sum_{n=1}^{4} i^n \cdot \|\alpha + i\beta\|^2$$
$$= \langle \alpha, \beta \rangle. \blacksquare$$

「Unitary Operator.

「Def. Let V be an Inn. Prod space,

An automorphism preserving inn, prod is called Unitary operator.

Equivalently, U is unitrary iff adjoint U^* exists and $U^* \circ U = U \circ U^* = I$.

Proof of the equivalence: Suppose that U is unitrary. Then, for any $\alpha, \beta \in V$,

$$\langle U(\alpha), \beta \rangle = \langle U(\alpha), (U \circ U^{-1})(\beta) \rangle = \langle \alpha, U^{-1}(\beta) \rangle.$$

So $U^{-1} = U^*$. Conversely, suppose U^* exists and $U^* = U^{-1}$. Then invertible is clear, and

$$\langle U(\alpha), U(\beta) \rangle = \langle \alpha, (U^* \circ U)(\beta) \rangle = \langle \alpha, \beta \rangle. \quad ☒$$

「Def. A real or Complex Matrix $A \in M_{n\times n}$ is called orthogonal if $A^t A = I$.

「Thm. For any $B \in GL_n(\mathbb{C})$, there exists a unique lower-triangular M with positive diagonal s.t. MB is unitary.

Proof. Since B is invertible, the row vectors $\{\beta_1, \cdots, \beta_n\}$ is basis for $\mathbb{C}^n$.

Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for $\mathbb{C}^n$ obtained from β_i's by Gram-Schmidt.

Now, for each $1 \le k \le n$, $\alpha_k = \beta_k - \sum_{j<k} \frac{\langle \beta_k, \alpha_j \rangle}{\|\alpha_j\|^2} \cdot \alpha_j$. And since $\operatorname{span}\{\beta_1, \cdots, \beta_{k-1}\} = \operatorname{span}\{\alpha_1, \cdots, \alpha_{k-1}\}$

$$\alpha_k = \beta_k - \sum_{j<k} C_{kj} \beta_j \quad \text{for some } \underset{\text{unique}}{C_{kj}} \in F.$$

Let $U = \begin{bmatrix} \frac{\alpha_1}{\|\alpha_1\|} & \cdots & \frac{\alpha_n}{\|\alpha_n\|} \end{bmatrix}$ and $M_{kj} = \begin{cases} -\frac{1}{\|\alpha_k\|} \cdot C_{kj} & \text{if } j<k \\ \frac{1}{\|\alpha_k\|} & \text{if } j=k \\ 0 & \text{otherwise.} \end{cases}$

↑ Revised in After page.

⌈Unique decomposition Theorem. (OR QR decomposition)

For any $B \in GL_n(\mathbb{C})$, there exists a unique lower-triangular M with positive diagonal s.t

MB is unitary.

Proof. Since B is invertible, the row vectors of B, $\{\beta_1, \cdots, \beta_n\}$ is basis for V.

Apply Gram-Schmidt process, that is, for each $1 \le k \le n$, put

$$\alpha_k = \beta_k - \sum_{k>j} \frac{\langle \beta_k, \alpha_j \rangle}{\|\alpha_j\|^2} \cdot \alpha_j$$

Then, $\mathrm{span}\{\alpha_1, \cdots, \alpha_r\} = \mathrm{span}\{\beta_1, \cdots, \beta_r\}$ by induction, and $\{\alpha_1, \cdots, \alpha_n\}$ is orthonormal.

Put $U = \begin{bmatrix} \frac{\alpha_1}{\|\alpha_1\|} & \cdots & \frac{\alpha_n}{\|\alpha_n\|} \end{bmatrix}$, then clearly $U^* \cdot U = I$, because $\{\alpha_1, \cdots, \alpha_n\}$ is orthogonal

And, for each $1 \le k \le n$, $\alpha_k = \beta_k - \sum_{k>j} C_{kj} \beta_j$ for some C_{kj}.

Put

$$M_{kj} = \begin{cases} -\frac{C_{kj}}{\|\alpha_k\|} & \text{if } k>j \\ \frac{1}{\|\alpha_k\|} & \text{if } k=j \\ 0 & \text{otherwise.} \end{cases}$$

Then, each $1 \le k \le n$, $U_k =$

$$\begin{pmatrix} \ddots & & 0 \\ & \ddots & \\ M_{kj} & & \ddots \end{pmatrix} \cdot \begin{pmatrix} \beta_1 & \cdots & \beta_n \end{pmatrix} = U$$

So $MB = U$. To show the uniqueness, denote

$T^+(n) = \{ M \in M_{n \times n}(\mathbb{C}) \mid M \text{ is lower triangle, positive diagonals.} \}$.

Let $M_1, M_2 \in T^+(n)$ satisfies M_1B, M_2B are unitary. Since unitary matrices forms a group,

$$(M_1B) \cdot (M_2B)^{-1} = M_1M_2^{-1} \text{ is also unitary.}$$

And, M_2^{-1} is lower-triangular with positive diagonals, so is $M_1M_2^{-1}$. Now,

$$(M_1M_2^{-1})^{-1} = (M_1M_2^{-1})^*, \text{ so } M_1M_2^{-1} \text{ is diagonal.}$$

So, $M_1M_2^{-1} = I$, or $M_1 = M_2$. ◻

⌈Corollary. For each $B \in GL(n)$, there exists unique matrix K, A, N s.t $B = KAN$,

where K is unitary, A is diagonal with positive values, N is upper-triangle with diagonals 1

+Note. Let $T_n(F)$ be the set of $n \times n$ upper-triangle matrices with all diagonals are 1.

Then, $(T_n(F), \cdot)$ is group, because; for any $A \in T_n(F)$, $A = I + N$,

where N is strictly upper triangular with $N^n = 0$, nilpotent. So,

$$A^{-1} = I - N + N^2 - N^3 + \cdots + (-N)^{n-1},$$

and $A^{-1} \in T_n(F)$ is clear.

$\boxed{QR}$ - Factorization.

Let $A = (w_1 \; \cdots \; w_n)$ be an invertible matrix.

And, for each $1 \le k \le n$, $\quad V_k = W_k - \sum_{i=1}^{k-1} \frac{\langle W_k, V_i \rangle}{\|V_i\|^2} \cdot V_i$.

Then, $\{V_1, \cdots, V_n\}$ is an orthogonal, so Put $U_k = V_k / \|V_k\|$, then $\{U_1, \cdots, U_n\}$ is an orthonormal. Clearly, $Q \overset{\text{def}}{=} (U_1 \; \cdots \; U_n)$ is Unitary.

Moreover, for each $1 \le k \le n$,

$$W_k = \|V_k\| \cdot U_k + \sum_{i=1}^{k-1} \langle W_k, U_i \rangle \cdot U_i.$$

Define

$$R_{jk} = \begin{cases} \|V_j\| & j = k \\ \langle W_k, U_j \rangle & j < k \\ 0 & j > k \end{cases}$$

Then, R is upper-triangular, and satisfying

$$Q^{-1}A = Q^{*}A = \begin{pmatrix} \overline{U_1} \\ \vdots \\ \overline{U_n} \end{pmatrix} \cdot \begin{pmatrix} W_1 & \cdots & W_n \end{pmatrix} = \begin{pmatrix} \langle W_1, U_1 \rangle & \langle W_2, U_1 \rangle & \cdots & \langle W_n, U_1 \rangle \\ \langle W_1, U_2 \rangle & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \langle W_1, U_n \rangle & \cdots & \cdots & \langle W_n, U_n \rangle \end{pmatrix}$$

$$= R.$$

So, $A = QR$.

「Matrix Decomposition.

In this page, our ambient space is $M_{n\times n}(F)$, simply M.

Given $A \in M$, denote $A^{[i]}$ is i'th row of A, $A_{[j]}$ is j'th Column of A.

With standard inner product on F^n, we can write

$$A \cdot B = \left(\left\langle A_{[i]}, B^{[j]} \right\rangle \right)_{i,j}.$$

and $$\left\langle A_{[i]}, B^{[j]} \right\rangle = \sum_{k=1}^{n} A_{ki} \cdot B_{ij}$$

「Cyclic subspace.

In this, every vector space is finite-dimensional, and $T: V \to V$ is arbitrary linear operator.

「Def. Given a vector $\alpha \in V$, define T-cyclic subspace generated by α is:

$$Z(\alpha; T) \overset{\text{def}}{=} \{ g(T)(\alpha) \mid g(t) \in F[t] \} = \operatorname{span}_F \{ T^k(\alpha) \mid k \geq 0 \}$$

If $Z(\alpha; T) = V$, then α is called a cyclic vector for T.

「Def. Given $\alpha \in V$, the kernel of a map

$$\varphi_T^{\alpha}: F[t] \to V : g(t) \mapsto g(T)(\alpha)$$

is called T-annihilator of α, and since $F[t]$ is P.I.D., it has unique monic generator.

Denote $\quad \ker \varphi_T^{\alpha} = M(\alpha; T) = (p_\alpha)$ where p_α is monic.

And, p_α divides the minimal polynomial of T.

「Thm. Given non-zero vector $\alpha \in V$, and T-annihilator of α, the followings hold;

1). $\dim Z(\alpha; T) = \deg p_\alpha$.

2). $Z(\alpha; T) = \operatorname{Span}_F \{ T^i(\alpha) \mid 0 \leq i \leq k-1 \}$ where $k = \deg p_\alpha$

3). The restriction $T|_{Z(\alpha;T)}$ has a minimal polynomial as p_α.

Prof. Let $g(t) \in F[t]$ be given. Then, by the division algorithm,

$$g = p_\alpha \cdot q + r \quad (r = 0 \text{ or } \deg r < \deg p_\alpha)$$

So, $g(T)(\alpha) = (p_\alpha \cdot q)(T)(\alpha) + r(T)(\alpha) = r(T)(\alpha) \in \operatorname{Span}_F \{ T^i(\alpha) \mid 0 \leq i \leq k-1 \}$.

And, these are linearly independent, by the minimality of p_α.

To show the last assertion, check that $Z(\alpha; T)$ is T-invariant. Now, given $g \in F[t]$,

$$(p_\alpha(T|_Z) \circ g(T))(\alpha) = (p_\alpha(T) \circ g(T))(\alpha) = g(T)(0) = 0.$$

And, by the minimality of p_α, it is also minimal polynomial of $T|_Z$.

「The Companion Matrix.

Let U be a linear operator on W of dimension $k \in \mathbb{N}$, which has a cyclic vector α.
Let $\alpha_i = U^{i-1}(\alpha)$, $i = 1, \cdots, k$. Then, $\{\alpha_1, \cdots, \alpha_k\}$ is ordered basis for U, and

$\alpha_{i+1} = U^i(\alpha) = U(\alpha_i)$, $i = 1, \cdots, k-1$, and for minimal polynomial p_α of U,

$$p_\alpha(U)(\alpha) = U^k(\alpha) + c_{k-1} \cdot U^{k-1}(\alpha) + \cdots + c_1 U(\alpha) + c_0 \alpha = 0.$$

So $U^k(\alpha) = U(\alpha_k) = -c_0 \alpha_1 - c_1 \alpha_2 - \cdots - c_{k-1} \cdot \alpha_k$. Now, for $\mathcal{B}$

$$[U]_{\mathcal{B}} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & & 0 & -c_1 \\ 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & 0 & \ddots & 0 & \vdots \\ 0 & 0 & \cdots & 1 & -c_{k-1} \end{bmatrix}$$

is called the Companion Matrix of p_α.

Def. Let V be an fin.dim inn. prod space.
A linear operator $T: V \to V$ is called Normal if $T \circ T^* = T^* \circ T$.
+ Note: the existence of adjoint is guaranteed by the Riesz Representation Theorem.

Thm. Let V be an fin.dim. inn.prod space, and T be a Hermitian.
Then, every eigenvalue of T is real, and eigenvectors of distinct eigenvalues are orthogonal.
Proof. Given eigenvalue c of T, for any eigenvector $\alpha \in V$ corresponding to c, observe

$$c \cdot \langle \alpha, \alpha \rangle \underset{\text{def}}{=} \langle c\alpha, \alpha \rangle \underset{T\alpha = c\alpha}{=} \langle T(\alpha), \alpha \rangle \underset{V \text{ fin. adj exist}}{=} \langle \alpha, T^*(\alpha) \rangle \underset{\text{Hermitian}}{=} \langle \alpha, T(\alpha) \rangle \underset{T\alpha = c\alpha}{=} \langle \alpha, c\alpha \rangle \underset{\text{def}}{=} \bar{c} \cdot \langle \alpha, \alpha \rangle$$

Since $\langle \alpha, \alpha \rangle$ is positive real number ($\because \alpha \neq 0$), so $c = \bar{c}$, or $c \in \mathbb{R}$.
Moreover, let $c_1, c_2 \in \mathbb{R}$ be distinct eigenvalues, and α_1, α_2 be eigenvectors of c_1, c_2, respectively. Then,

$$c_1 \cdot \langle \alpha_1, \alpha_2 \rangle = \langle c_1\alpha_1, \alpha_2 \rangle = \langle T(\alpha_1), \alpha_2 \rangle = \langle \alpha_1, T^*(\alpha_2) \rangle = \langle \alpha_1, T(\alpha_2) \rangle = \langle \alpha_1, c_2\alpha_2 \rangle = \bar{c}_2 \langle \alpha_1, \alpha_2 \rangle$$

By the first assertion, $\bar{c}_2 = c_2$, so $(c_1 - c_2) \cdot \langle \alpha_1, \alpha_2 \rangle = 0$.
Finally, $\mathbb{C}$ has no zero divisor, $\langle \alpha_1, \alpha_2 \rangle = 0$. (A kind of joke.)

Thm. Every Hermitian on fin.dim. inn. prod space has an eigenvalue.
Proof. Set V be a such space, and T is Hermitian. Take $\mathcal{B}$ be an orthonormal basis for V.
Let $A = [T]_{\mathcal{B}}$. Then, $A = [T^*]_{\mathcal{B}} = [T]_{\mathcal{B}}^* = A^*$.
Put $W \overset{\text{def}}{=} \{X \in M_{n \times 1}(\mathbb{C})\}$, with inner product $\langle X, Y \rangle = Y^* X$.
Then, a map

$$H: W \to W : X \mapsto AX$$

is Hermitian. Moreover, the charateristic polynomial of H:

$$\det(xI - H) = \det(xI - A)$$

is a polynomial of degree n over $\mathbb{C}$, so it has a root $c \in \mathbb{C}$.
Thus, c is an eigenvalue of H, Hermitian, so it is a real number.

Note: Characterstic Polynomial of Hermitian splits over the given field $\mathbb{R}$ or $\mathbb{C}$.

┌Lemma. Let V be an fin. dim inn prod space, and T be a linear operator.
For any T-invariant subspace $W<V$, the orthogonal complement $W^\perp$ is T^*-invariant.

Proof. $W^\perp$ is T^*-invariant iff for any $\alpha\in W^\perp$, $T(\alpha)\in W^\perp$

Given $\alpha\in W^\perp$, $\langle\alpha, T(\beta)\rangle=0$ for all $\beta\in W$, since $T(\beta)\in W$.
That is, $\langle T^*(\alpha),\beta\rangle=0$ for all $\beta\in W$, or $T^*(\alpha)\in W^\perp$.

┌Thm. Let V be an fin. dim inn prod space, and let $T: V\to V$ be a linear operator.
T is Hermitian if and only if there is an orthonormal basis for V consisted by eigenvectors of T.

Proof. Using induction for $\dim V=n$.
Since T is Hermitian, it has at least one eigenvector $\alpha_1\in V$. Now,
$\{\alpha_1/\|\alpha_1\|\}$ is the orthonormal basis we desired. Now, let the statement hold for $n-1$.
Let $\dim V=n$. Then,